HAW HAMBURG

IE6, OSL – Lab Test 2

Management of processes, process priorities and concurrent access to resources is essential for all modern operating systems.

Learning target

- Knowledge about processes and priorities.
- Knowledge about locking of concurrent accesses.

During this lab, we will only use a standard office PC with Linux OS.

Working time: 3 hours.

2.1 Lab test „scheduling“

Develop a C program to generate two competing processes, both writing characters to the screen by using an appropriate output function. Watch the output and explain it regarding the properties of the output functions.

a) Change the „nice value“ of one process by using the **`renice`** command. What's the effect to the output? Explain!
b) Insert a waiting period of 1 ms after each output. What's the effect to the assignment of calculation time?
c) Run a very time consuming program simultaneously. How does this affect the results of a) and b)?

Note: Stream functions are buffering the output and this affects the scheduler's decisions.

2.2 Lab test "Concurrent access to a file"

Preparation: Write a C program running two processes A and B continuously:

1. Process A writes a short text to a file.
2. Then process B has to read this file.
3. After this write-/read-cycle, process A has to erase the file,
4. then the processes change roles: B writes to the file, A reads from the file.

Lab tasks to manage concurrent file access:

a) Use a lock file first to implement the exclusive usage of the file.
b) Implement an exclusive access to the file by usage of **`fcntl()`.**
Study the manual first.

Check if your program runs correctly. Compare both attempts. Name the advantages and disadvantages of each method.